

Connected component analysis

- Also known as connected component labeling
- Connected pieces of true values in a binary image can be used to extract useful information
- Mostly used in computer vision but we will look into this topic a bit
- Simplest connected component analysis is number of connected pixels
- There two different neighborhoods to decide which pixels are connected: von Neumann neighborhood (4-connected) and Moore neighborhood (8-connected)
- There are many algorithms to find connected components, most common and simple methods are named as one-pass and two-pass algorithms.
- One pass algorithm is the simplest one
 - Starting from one corner of the image, search every pixel one by one.
 - If an unlabeled true pixel is found, start a new label or component
 - To detect all pixels in this new label, label and push the location of the first pixel into a stack
 - While there are pixels in the stack
 - * Take a pixel location from the stack
 - * For every true valued neighbor of the pixel, mark the neighbor with the current label, and push its location to the stack
- Once connected components are labeled, there many methods to analyze their shape
- Simplest of these methods is moments (and moment invariants)
 - Raw moments are relatively useful. For instance: area of the shape (M_{00}), centroid ($\frac{M_{10}}{M_{00}}, \frac{M_{01}}{M_{00}}$)
 - Raw moments are very easy to calculate $M_{ij} = \sum_x \sum_y x^i y^j I_{x,y}$
 - The formulas for moment invariants are complex, but they are mathematical formulas (i.e., lots of typing)
 - Moment invariants are invariant to translation (movement), scale, and rotation. Therefore, the values obtained from them will be closer if the shapes are similar. There are 7 well-known moment invariants (up to 3^{rd} order), though there is another 3^{rd} order moment invariant which is lesser known.
- Another useful information that can be extracted from the connected components is the contour
 - Contours is a one dimensional time series with two dimensional data: time vs. (x, y)
 - There are many methods to extract contours: Moore neighbor tracing, Radial sweep, Pavlidis' algorithm are some examples to these

- Once the contours are extracted, the relationship between contour points and their relation to the overall shape can be used for many methods
- Simplest of these methods is perimeter to square root of area ratio

Morphological operations

- These operations work on binary images to modify them for some purpose
- Simplest and most commonly used morphological operation in image processing is to reduce noise in mask images
- We will see methods to reduce the noise, adjust the size of the components and few methods to assist computer vision tasks

Dilation, erosion

- These methods are used to increase (dilation) or reduce (erosion) the true regions in the binary image
- Similar to convolution, these filters require a kernel
- This kernel is binary in nature
- Dilation operation can be remembered as At least one
 - Center the kernel on the pixel to be determined
 - If any of the kernel values that are true is hit a true pixel in the image
 - Set the pixel to true
 - Otherwise set it to false
- Erosion operation can be remembered as All
 - Center the kernel on the pixel to be determined
 - If all of the kernel values that are true is hit to a true pixel in the image
 - Set the pixel to true
 - Otherwise set the pixel value to false
- Often a box kernel is used, though it is better to use a circular kernel if the size is large

Opening, closing

- Opening clears small noisy components from the image (opening as it open gaps)
- Closing clears holes in components
- Opening is erosion followed by dilation
- Closing is dilation followed by erosion

Simple operations

- Negation (not): simply set true pixels to false and false pixels to true
- Or/and/xor: These operations require two images, simply perform the boolean operation by taking pixels of the same coordinates in these images. For instance, and operation can be used to refine thresholding in one color component with the thresholding in another (joining hue and saturation)

Thinning, skeletonization

- These methods are used to reduce a shape into connected lines
- Thinning is often used to clear detected edges in an image (edge detection → threshold → thinning)
- Most common methods to perform thinning is to use gradient descent (for grayscale images) and hit-and-miss transform (for binary images)
- There are different skeletonization algorithms which have similar results. Repeated thinning can be used to skeletonize an image. Medial axis transform is another similar method.

Thickening, convex hull

- These operations remove concave parts of the image
- Thickening is performed using hit-and-miss transform
- Convex hull is calculated over contours

Hit-and-miss transform

- Similar to erosion
- Instead of simple true or false, there are three kernel states: true (1), false (-1), doesn't matter (0)
- In this method, kernel values that are true should hit to true pixels and the ones that are false should hit to false pixels in order for the new value to be true
- For instance, following kernel detects north west corners

0	1	0
1	1	-1
0	-1	-1